Guillermo Fernando Farfán Romero

Senior Database Administrator (DBA) & Data Engineer

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PROFESSIONAL SUMMARY

Senior Database Administrator (DBA) and Data Engineer with 8+ years of experience architecting, administering, optimizing, and scaling relational, NoSQL, vector, and high-throughput Data Warehouse platforms. Specialist in **Microsoft SQL Server**, **PostgreSQL (with pgvector for semantic search/AI)**, advanced **Performance Tuning** (execution plan analysis, table partitioning, deep indexing, deadlock resolution), cloud data architectures on **Google Cloud Platform (Cloud SQL, BigQuery)**, and automated ELT/ETL pipelines with **Apache Airflow, dbt, and Python (Polars, PyArrow, SQLAlchemy)**.

Combines deep technical expertise in data persistence, governance, and business continuity (DRP/BCP with strict RPO/RTO) with strategic team leadership, backed by a background in Information Systems and advanced coursework in Labor Relations (UBA).

TECHNICAL SKILLS & COMPETENCIES

- **Database Engines (RDBMS & Vector):** Microsoft SQL Server (2012-2022, AlwaysOn Availability Groups, SSIS, SSRS, T-SQL, CDC), PostgreSQL (12-16, pgvector, Declarative Partitioning, Streaming Replication, PgBouncer, Connection Pooling), MySQL, MariaDB.
- **Data Warehousing, Lakehouse & NoSQL:** Google BigQuery (Partitioning, Clustering, Slot Optimization, Star Schema / Kimball, BigQuery Vector Search), Redis (Distributed Caching, Clustering, Pub-Sub, In-Memory Storage), Google Cloud Pub/Sub, MongoDB, Cloud Firestore.
- **Data Engineering & Orchestration:** Apache Airflow, Python (Polars, PyArrow, Pandas, SQLAlchemy 2.0, PyODBC), Alembic (Zero-Downtime Migrations), dbt (Data Modeling & Testing), Data Contracts, OpenLineage, Data Quality, Batch & Streaming ETL/ELT Pipelines, Celery, Bash Scripting.
- **Infrastructure, SAN & Storage:** Google Cloud Platform (Cloud SQL, BigQuery, Persistent Disk), SAN Storage (HPE Alletra / 3PAR, MPIO Round Robin, CSV), Microsoft Hyper-V 2025 / VMware, Veeam Backup & Replication (VBR), Linux (SLES 15 SP7, Ubuntu), XFS, LVM, IOPS & Throughput Tuning.
- **Diagnostics & Performance Tuning:** Query Execution Plans, SQL Profiler, Extended Events, Index Fragmentation Management, Slow Query Logs, Deadlock/Blocking Resolution, Real-time Observability with IBM Instana APM, Zabbix, and OpenTelemetry.
- **Security & Business Continuity:** Transactional Replication, Automatic Failover, Strict RPO/RTO Backup/Restore Strategies, TDE/SSL Encryption, Role-Based Access Control (RBAC), Data Governance.

PROFESSIONAL EXPERIENCE

Technical Lead & Senior Systems Engineer (Data Architecture Focus) | Reforest Latam

May 2026 – Present | Full-time (Remote)

- **Architected and implemented a distributed caching and in-memory persistence layer with Redis** for environmental telemetry platforms, driving a **45% speedup in concurrent analytical and transactional queries**.
- **Standardized 100% of database lifecycle and schema version control**, establishing automated schema migrations with **Alembic** and CI/CD pipelines for **zero-downtime deployments** without service interruptions.
- **Led relational persistence design and governance in PostgreSQL** for high-concurrency platforms, enforcing referential integrity, declarative table partitioning, and strategic normalization.
- **Directed and mentored an engineering team of 6**, defining SQL query optimization guidelines, complex transaction tuning, and schema decoupling standards.

Senior Data Infrastructure Consultant & DBA | Independent Consulting & Enterprise Projects

January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)

- **Architected high-throughput storage and database persistence on a Microsoft Hyper-V 2025 cluster (Universal Group)** with HPE Alletra/3PAR SAN storage (55 TB CSV, MPIO), orchestrating replication and backup with **Veeam Backup & Replication (VBR)**.
- **Led database and VM migration to GCP (Grupo Ramos - Phase II)** utilizing **Migrate to Virtual Machines (M4CE)** and Cloud SQL, applying compute sizing and FinOps policies for high-traffic database workloads.
- **Engineered and deployed automated ELT pipelines with Apache Airflow and Python (Polars, PyArrow, SQLAlchemy)**, cutting **batch windows by 60%** across **+500 GB daily** datasets in **Google BigQuery** (Star Schema, clustering).
- **Maximized storage I/O throughput and reduced latency** across Linux servers (SLES 15 SP7 / Ubuntu) and Cloud SQL through block-aligned XFS and advanced LVM management.
- **Implemented real-time transactional observability with IBM Instana APM, Zabbix, and OpenTelemetry**, tracking TPS, p95/p99 query latencies, and eliminating connection pool bottlenecks.

Senior Software Engineer & Database Administrator | University of Buenos Aires (UBA)
March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)

- **Spearheaded comprehensive performance tuning across 8+ TB of mission-critical data** (+50,000 active users), restructuring indexes, optimizing execution plans, and resolving deadlocks to **cut p95 query latency by 55%**.
- **Guaranteed RPO < 15 min and RTO < 1 hour** through High Availability (HA) replication architectures and robust Disaster Recovery Plans (DRP/BCP).
- **Developed automated Python ETL pipelines**, accelerating data cleansing, validation, and migration from legacy databases to modern relational schemas.

Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)
January 2020 – December 2023 | Full-time (Buenos Aires, Argentina)

- **Optimized maintenance schedules, native compression, and differential backup rotation on SQL Server and PostgreSQL**, achieving a **35% reduction in overnight backup windows**.
- **Maintained 99.9% operational database uptime (SLA)** across municipal citizen portals serving 100,000+ monthly users.
- **Implemented data access governance and auditing**, establishing historical data retention policies and strict information security compliance.

EDUCATION

- **Associate Degree in Information Systems** | National Technological University (UTN) *(In Progress — Expected Graduation: December 2027)*
- **Bachelor's Studies in Labor Relations & Human Resources** | University of Buenos Aires (UBA) *(Advanced University Coursework Completed — Focus on Organizational Leadership, Talent Strategy & Engineering Team Dynamics)*

KEY CERTIFICATIONS *(100% VERIFIABLE ON CREDLY)*

- **Professional Data & Backend Certifications (Credly):** Google Data Analytics Professional Certificate | Meta Back-End Developer Certificate | IBM DevOps and Software Engineering Professional Certificate.
- **Cloud & Persistence Specializations:** Google Cloud Fundamentals: Core Infrastructure (Google Cloud) | IBM Full Stack Software Developer Certificate | Oracle Next Education F2 Back-End.

LANGUAGES

- **Spanish:** Native
- **English:** Professional Working Proficiency (B2 — Fluent technical communication for live engineering syncs, RFCs, technical documentation, and cross-functional collaboration)